

DRAWING LIST FOR TEST PACKAGE

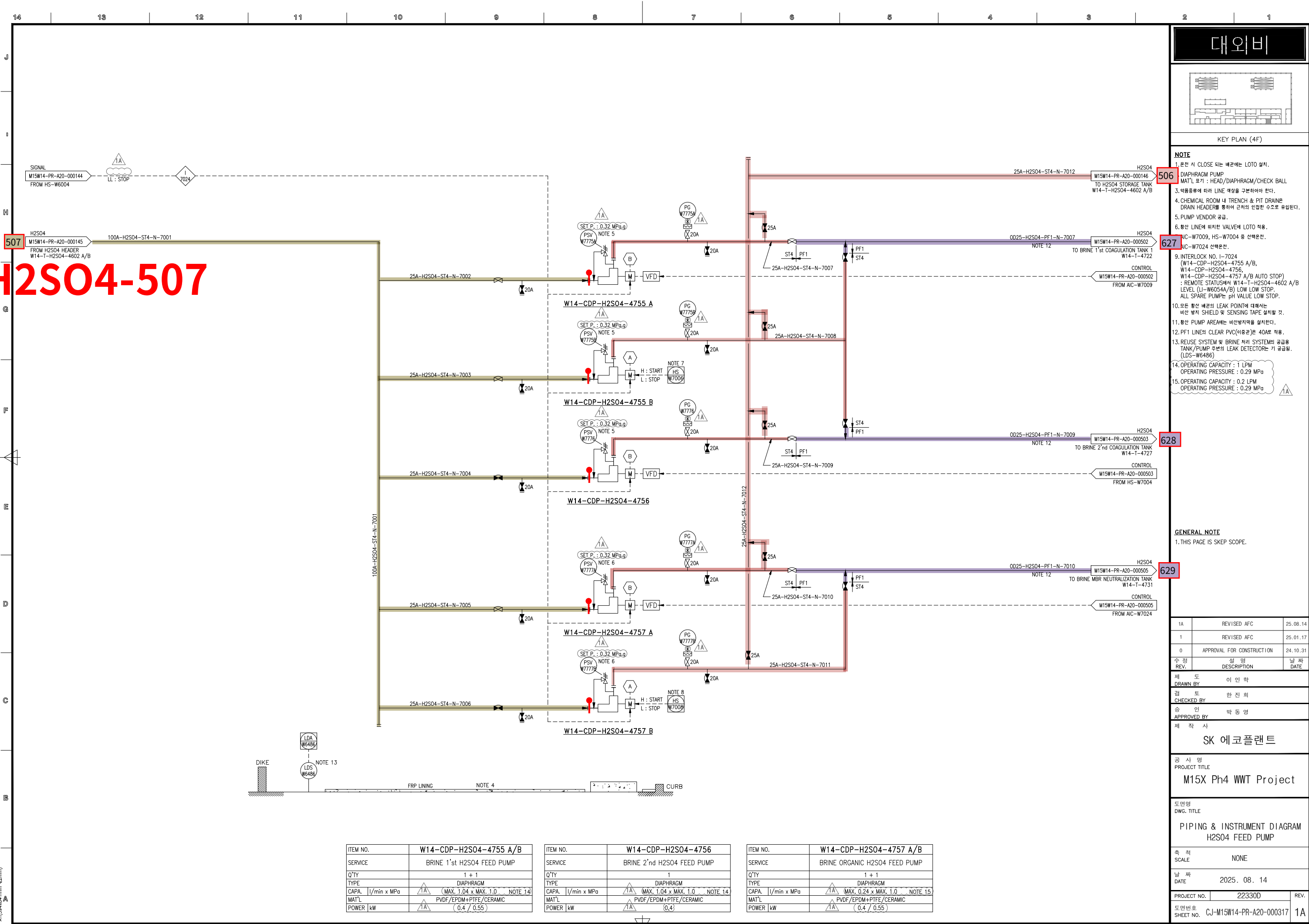
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

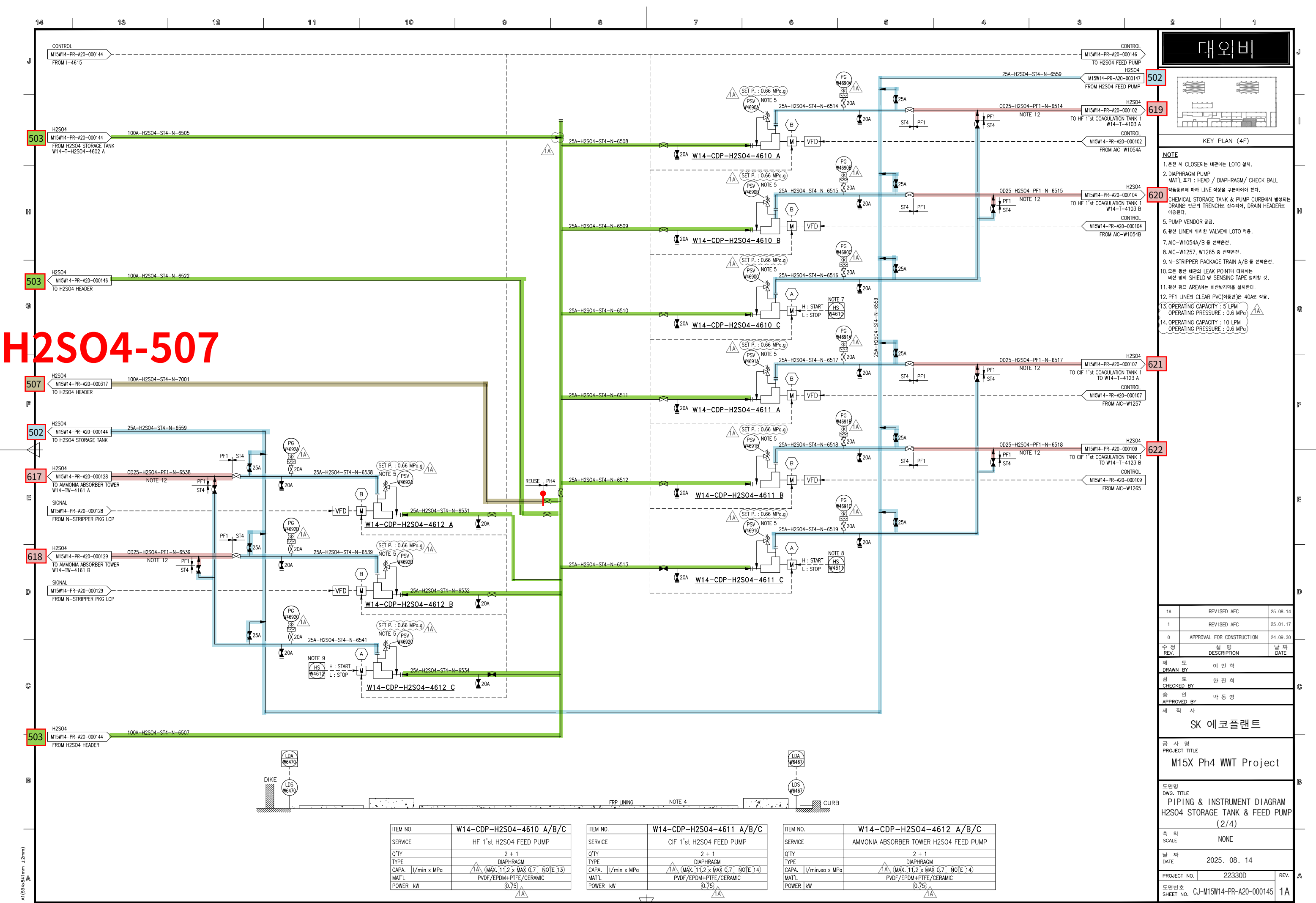
H2SO4-507

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	507	4FA	25A-H2SO4-ST4-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
2	507	4FA	25A-H2SO4-ST4-N-7003-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
3	507	4FA	25A-H2SO4-ST4-N-7004-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
4	507	4FA	25A-H2SO4-ST4-N-7005-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
5	507	4FA	25A-H2SO4-ST4-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
6	507	4FA	100A-H2SO4-ST4-N-7001-01	1 OF 1	0	CJ-M15W14-PR-A20-000145/317		

H2SO4-507



H2SO4-507



대외비

KEY PLAN (4F)

NOTE

1. 운전 시 CLOSE되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP

MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL

작동종류에 따라 LINE 색상을 구분하여야 한다.

3. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 침수되어, DRAIN HEADER로 이송된다.

5. PUMP VENDOR 공급.

6. 황산 LINE에 위치한 VALVE에 LOTO 적용.

7. AIC-W1054A/B 중 선택운전.

8. AIC-W1257, W1265 중 선택운전.

9. N-STRIPPER PACKAGE TRAIN A/B 중 선택운전.

10. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.

11. 황산 침트 AREA에는 비산방지막을 설치한다.

12. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.

13. OPERATING CAPACITY : 5 LPM

OPERATING PRESSURE : 0.6 MPa

14. OPERATING CAPACITY : 10 LPM

OPERATING PRESSURE : 0.6 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE
제도 DRAWN BY	이인학	
검토 CHECKED BY	한진희	
승인 APPROVED BY	박동영	
제작사	SK 에코플랜트	
공사명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM	
	H2SO4 STORAGE TANK & FEED PUMP	
속척 SCALE	NONE	
날짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145	1A



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* REDUCE: TR	* SUB-IN: STU	* REINFORCING PAD: RP	* CONJUG: RC	* ECC RED: RE	* CONSUMING: RCS/RCS1	* ECC SANGING: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* SUB-IN: STU	* SOCKET(HOPE/PC): SC	* SHOULDER(HOPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VAL: VG/VG1/VG2	* BALL VAL: VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VAL: VF/VF4	* CHECK VAL: VC/VC1/VC2/VCS/VCA/VCS/VCA/VCT					
* GLOBE VAL: VL/VL1/VL2	* DIAPHRAGM VAL: VD/VD0	* DAMPER: VP	* COUPLING COUPLING: SOC	* FLEXIBLE JOINT(STEEL): FLX	* FLEXIBLE JOINT(RUBBER): FLXR	* FLEXIBLE JOINT(STEEL PIPE LINING): FLXL	* FLEXIBLE JOINT: FLX	* FLEXIBLE JOINT: FLX	* FLEXIBLE JOINT: FLX	* FLEXIBLE JOINT: FLX	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH		
* STRAIGHT TYPE: YSTR	* STRAIGHT TYPE: YSTR	* STRAINER(BASKET TYPE): BSTR	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT: SB	* BOLT: BL		

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CLIENT	 1	
CONTRACTOR		PROJECT NAME M15X Ph4 Project
COMMODITY	H2S04	JOB NO. 22330D
LINE NO.	25A-H2S04-ST4-N-7002	SH'T NO. REV. 01 / 01 0

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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

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<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

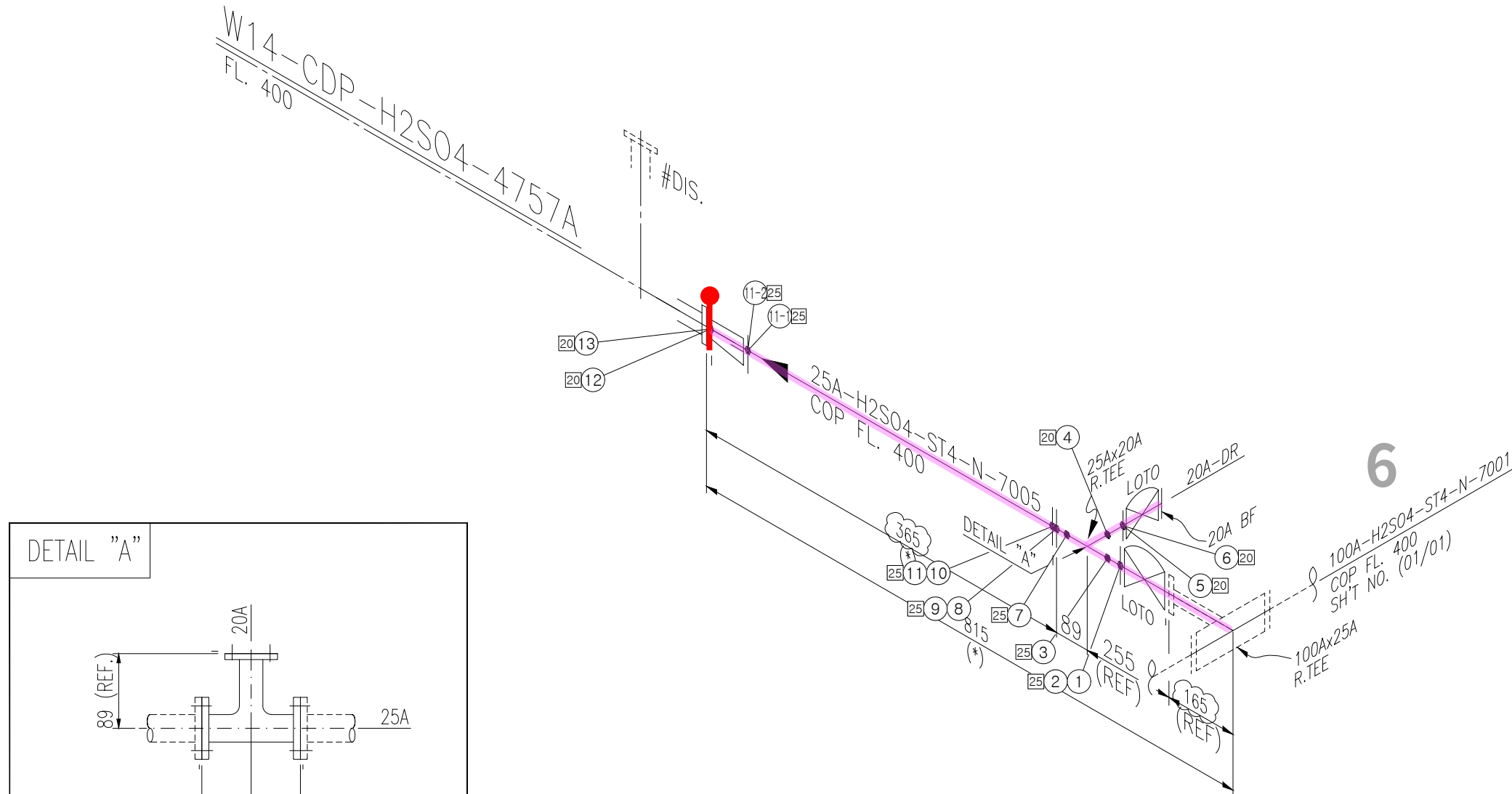
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000317							REV. NO.	DATE	DESCRIPTION	BY

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CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY H2S04		JOB NO. 22330D	
LINE NO. 25A-H2S04-ST4-N-7005		SH'T NO. 01 / 01	REV. 0



<NOTE>
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배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	1.0	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동호
						REF. P & ID CJ-M15W14-PR-A20-000317						



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상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.



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2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR 	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	4FA									M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID	CJ-M15W14-PR-A20-000317							REV. NO.	DATE	DESCRIPTION	BY
																01 / 01	0

